PEST MANAGEMENT DECISION GUIDE: GREEN AND YELLOW LIST

Late blight on Tomato

Phytophthora infestans Late blight



Late blight symptoms on tomato (Photo from: www.organicgardening.com)



Symptoms on tomato plant stem (Photo by: Elizabeth Bush, Virginia Polytechnic Institute and State University, Bugwood.org).

Prevention

- Transplant disease-free seedlings. Plant certified seed of approved resistant varieties like Yara, Rodade, 4 star 9030.
- Plant in basins or apply mulch to decrease the need for watering.
- Plant tomato plants at 60cm by 60cm intervals to allow air flow between plants.
- Use drip irrigation instead of overhead irrigation to prevent leaf wetness.
- Crop rotation - avoid planting on sites that were previously planted with Solanaceae family crops like potatoes and eggplants. Rotate with cereals like maize.
- Destroy all tomato debris after harvest by burying at around 1m deep (ensure complete coverage).

Monitoring

- Monitor twice a week during the dry season for symptoms and on a daily basis during the rainy season.
- Apply control measures once you see one plant showing symptoms.
- Check for plants or fruits touching or leaning on the ground.
- Search for dark, greyish lesions on leaves.
- Check stems to see if they have brown to black areas, plants dry up and collapse quickly.
- Look for fuzzy white fungal growth on the undersides of the leaves during humid weather.

Direct Control

- There are few direct control options for this disease.
- Use Chilli mixed with Muzwezi (Pencil bush) to treat. This is after pounding and drying Muzwezi then add chilli powder and soak in water. See factsheet for details.
- Use Neem oil spray (See factsheet for details) - can help reduce and prevent late blight.
- For early symptoms on plants, remove the infected plant and destroy it completely by burying up to 1m deep.

Direct Control / Restrictions

Direct Control	Restrictions
• Refer to product label for instructions.	
• Mancozeb-based products (DITHANE; ETHANE M45 80%WP; DIPACK; ROLIM 700WP; and others). Broad-spectrum contact fungicide with protective action. Multi-side action pesticide group.	• WHO toxicity class U (Unlikely to present acute hazard in normal use). Max 2 sprays with min 7 to 10 d retreatment interval, reentry interval (r.e.i. 3 d). Avoid treating wet crops and those suffering from drought. Do not spray near water.
• Copper-based products (copper salts, copper oxychlorides). Multi-side action pesticide group. Usually applied at 40-80 mg /20 litres, but double-check product labels. Never spray more than 6 kg copper per ha per year.	• WHO toxicity class II (moderately hazardous); For r.e.i. and max number of sprays refer to label instructions.When used too much, copper(II)-ions can accumulate in soil and water. It can burn leaves. Do not use during fruit set.
• Mancozeb and metalaxyl mixtures are available for resistance management (METALIN VICTORY; DILAX; and others). Nucleic acid synthesis pesticides.	• WHO toxicity class II (moderately hazardous); Max 3 sprays with min 10 to 14 d retreatment interval, For r.e.i. after spray and max number of sprays refer to label instructions.

Zambia

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